

CS 549: Performance Analysis of Computer Networks

Course Code	CS 549
Course Name	Performance Analysis of Computer Networks
L-T-P-C	3-0-0-3
Prerequisites	IC 210 Probability, statistics and random variables or equivalent, CS 310 Introduction to Communicating and Distributed Processes or consent of the instructor.
Intended for	BTech CSE & EE, MTech/MS/PhD in the area.
Distribution	Discipline elective for BTech CSE and EE, free elective for others
Approval	16th BoA (Feb 2017); 12th Senate (Mar 2017)

Course Contents

- **Introduction to computer networks:** A brief history and introduction to the Internet, Review of networking and layering. [3 Lectures]
- **Multiaccess communication and its analysis:** Slotted multi-access and the ALOHA system, Carrier sensing multiple access slotted ALOHA, Local Area Networks: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) and Ethernet: Slotted CSMA/CD, unslotted CSMA/CD and IEEE 802 standards, Link scheduling and Network capacity. [9 Lectures]
- **Routing algorithms and their analysis:** Algorithms for shortest path routing - Dijkstra's Algorithm, Bellman-Ford Algorithm and Generalized Dijkstras Algorithm, Optimal routing. [8 Lectures]
- **Queueing theory:** Brief review of Random processes, Introduction to Markov chains and queueing theory, traffic models, deterministic and stochastic analysis; Delay modeling using queueing theory Little's law, M/M/1, M/M/m, M/M/m/m, M/G/1 queueing systems, priority queueing. [10 Lectures]
- **Measurement techniques and experiment design:** Workloads, Monitors, Experiment design - 2k, 2kr factorial designs. [6 Lectures]
- **Advanced topics:** Network economics, Mobile IP, Multimedia streaming, VoIP, Content Distribution Networks, Software-defined networking and network function virtualization. [4 Lectures]

Text Books

1. D. Bertsekas and R.G. Gallager, **Data networks**, Prentice-hall, 1992.
2. Anurag Kumar, D. Manjunath and Joy Kuri, **Communication networking: an analytical approach**, Elsevier 2004.
3. J.F. Kurose and K.F. Ross, **Computer networking: a top-down approach**, Pearson, 2010.

Additional References

1. R. Jain, **The art of computer systems performance analysis**, Wiley India, 2013.
2. K.S. Trivedi, **Probability and Statistics with Reliability, Queueing, and Computer Science Applications**, John Wiley and Sons Ltd, 2016.
3. G. Varghese, **Network algorithms: an interdisciplinary approach to designing fast networked devices**, Morgan Kaufmann, 2004.
4. B. Hajek, **Notes for ECE 567 Communication network analysis**, University of Illinois, 2006. <http://www.ifp.illinois.edu/~hajek/Papers/networkanalysis.html>